Hsing-Hao Lee Curriculum Vitae

Contact Information

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Current position

2026–	Postdoctoral Research Associate Department of Psychology, New York University, USA Advisor: Dr. Clayton Curtis

Education

2021– 2026	Ph.D., Psychology (Cognition and Perception, Quantitative Concentration)
2021– 2025	Master of Philosophy, Psychology Department of Psychology, New York University, USA Advisor: Dr. Marisa Carrasco
2017– 2019	Master of Science, Psychology Department of Psychology, National Taiwan University, Taiwan Advisor: Dr. Su-Ling Yeh
2013– 2017	Bachelor of Science, Psychology Department of Psychology, National Cheng Kung University, Taiwan Advisor: Dr. Shulan Hsieh

Research Interests

1. **Attention:** How does visual attention modulate perception?
2. **Cognitive control:** How do people suppress distractions while maintaining relevant information?
3. **Working memory:** How do top-down factors affect visual working memory representations?
4. **Consciousness:** What are the differences between conscious and unconscious mental processes?

Publications and Manuscripts

1. **Lee, H. H.** & Carrasco, M. (preprint). Endogenous attention is invariant to sequential effects in performance around the visual field. *bioRxiv*.
2. **Lee, H. H.**, & Carrasco, M. (2026). Covert spatial attention is uniform across cardinal meridians despite differential adaptation. *Journal of Vision, 26*, 15.

3. Yeh, S. L., Li, S. H., **Lee, H. H.**, Kao, Y. W., Goh, J. O. S., Jingling, L., & Tsai, A. C. H. (2026). Cortical representation of apparent motion sequences in younger and older Adults: An fMRI study. *Scientific Reports, 16*, 26137.
4. Chen, Q., **Lee, H. H.**, Hanning, N.M., & Carrasco, M. (*preprint*). Transcranial magnetic stimulation to frontal cortex, unlike occipital stimulation, does not disrupt exogenous attention. *bioRxiv*.
5. **Lee, H. H.**, & Carrasco, M. (2025). Visual adaptation stronger at horizontal than vertical meridian: Linking performance with V1 cortical surface area. *Proceedings of the National Academy of Sciences of the United States of America, 122,* (29) e2507810122.
6. **Lee, H. H.** (2025). Intrinsically photosensitive retinal ganglion cells and visual processing: ipRGCs beyond non-image-forming functions. *Frontiers in Neuroscience, 19:*1635101.
7. **Lee, H. H.**, Fernández, A., & Carrasco, M. (2024). Adaptation and exogenous attention interact in the early visual cortex: A TMS study. *iScience, 27*(11), 111155.
8. **Lee, H. H.**, Groves, K., Ripollés, P., & Carrasco, M. (2024). Audiovisual integration in the McGurk effect is impervious to music training. *Scientific Reports, 14*, 3262.
9. **Lee, H. H.**, Liu, G. K., Chen, Y. C., & Yeh, S. L. (2024). Exploring quantitative measures in metacognition of emotion. *Scientific Reports, 14*, 1990.
10. Cheng, T., Chiu, L., Huang, L., Lin, Y. T., **Lee, H. H.**, Chen, Y. C., & Yeh, S. L. (2023). Predictive processing in the "Second Brain": From gut complex to meta-awareness. In *Expected Experiences* (pp. 170-194). Routledge.
11. **Lee, H. H.**, Chien, S. E., Lin, V., & Yeh, S. L. (2022). Seeing food fast and slow: Arousing pictures and words have reverse priorities in assessing awareness. *Cognition, 225*, 105144.
12. Chen, Y. T., **Lee, H. H.**, Shih, C. Y., Chen, Z. L., Beh, W. K., Yeh, S. L., & Wu, A. Y. (2022). An effective entropy-assisted mind-wandering detection system with EEG signals of MM-SART Database. *IEEE Journal of Biomedical and Health Informatics, 26*(8)*,* 3649-3660*.*
13. **Lee, H. H.,** Chen, Z. L., Yeh, S. L., Hsiao, J. H., & Wu, A. Y. (2021). When eyes wander around: Mind-wandering as revealed by eye movement analysis with hidden Markov models. *Sensors, 21,* 7569.
14. **Lee, H. H.**, Tu, Y. C., & Yeh, S. L. (2021) In search of blue-light effects on cognitive control. *Scientific Reports, 11*, 15505.
15. **Lee, H. H.,** & Yeh, S. L. (2021). Blue-light effects on saccadic eye movements and attentional disengagement. *Attention, Perception, & Psychophysics. 83,* 1713-1728.
16. Chien, S. E., Chu, L., **Lee, H. H.**, Yang, C. C., Lin, F. H., Yang, P. L., Wang, T. M., & Yeh, S. L. (2019). Age difference in perceived ease of use, curiosity, and implicit negative attitude toward robots. *ACM Transactions on Human-Robot Interaction (THRI)*, *8*(2):9.
17. **Lee, H. H.,** & Hsieh, S. (2017). Resting-state fMRI associated with stop-signal task performance in healthy middle-aged and elderly people. *Frontiers in Psychology*, *8*:766.

Manuscripts in preparation

1. **Lee, H. H.** & Carrasco, M. (*in preparation*). Does early visual cortex play a causal role in temporal attention? An TMS study.
2. **Lee, H. H.**, Kao, Y. W., Goh, J. O. S., & Yeh, S. L. (*in preparation*). Age-related differences in facial expression predictions: An fMRI study.

Selected Conference Presentations

1. **Lee, H. H.** & Carrasco, M. (2026). *Early visual cortex is not functionally necessary for temporal attention: A TMS study.* Poster presented at the Annual Meeting of the Society for Neuroscience. Washington, D.C.
2. **Lee, H. H.** & Carrasco, M. (2026). *Endogenous and exogenous attention remain uniform across cardinal meridians despite differential adaptation*. Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
3. Carrasco, M., Hanning, N. M., Fernández, A., Chen, Q., & **Lee, H. H.** (2026). *Cortical double dissociation for endogenous and exogenous covert spatial attention*. Talk presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
4. Chen, Q., **Lee, H. H.**, Fernández, A., Hanning, N. M., & Carrasco, M. (2025). *Neurostimulation reveals prefrontal cortex plays a central role in endogenous but not exogenous attention*. Poster presented at the Annual Meeting of the Society for Neuroscience 2025, San Diego, CA.
5. Carrasco, M., Chen, Q., **Lee, H. H.,** Fernández, A., & Hanning N. M. (2025). *TMS to the prefrontal cortex disrupts endogenous, but not exogenous, attention.* Talk presented at the Annual Meeting of the European Conference on Visual Perception, Mainz, Germany.
6. **Lee, H. H.**, & Carrasco, M. (2025). *Endogenous attention enhances contrast sensitivity similarly around cardinal meridians despite differential adaptation effects.* Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
7. Chen, Q., **Lee, H. H.,** Hoxha, K., Fernández, A., Hanning, N. M., & Carrasco, M. (2025). *Does human right frontal eye field (rFEF+) play a critical role in exogenous attention? A Transcranial Magnetic Stimulation (TMS) study*. Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
8. **Lee, H. H.** & Carrasco, M. (2024). *Visual adaptation is more pronounced at the horizontal than vertical meridian.* Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
9. **Lee, H. H.**, Fernandéz, A., & Carrasco, M. (2023). *Adaptation modulates the effect of covert exogenous attention in early visual cortex – A TMS study.* Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
10. **Lee, H. H.**, Liu, G. K., & Yeh, S. L. (2021) *I know I'm happy, and I'm right: Metacognition of emotion*. Poster presented at the Annual Meeting of European Conference on Visual Perception, Virtual Meeting.
11. Cheng, T., Chiu, L., Huang, L. T., Lin, Y. T., Chen, Y. C., **Lee. H. H.**, & Yeh, S. L. (2021) Metacognitive

confidence in perception, interoception, and memory. Poster presented at the Conference of Metacognition: New developments and challenges, Virtual Meeting.

12. **Lee, H. H.**, Chien, S. E., Lin, V., & Yeh, S. L. (2021*). Which comes first? Examining breaking continuous flash suppression time of high-calorie and low-calorie foods*. Poster presented at the Annual Meeting of the Vision Sciences Society, Virtual meeting.
13. Chen, Z. L., **Lee, H. H.**, Chen, Y. T., Wu, A. Y., & Yeh, S. L. (2021). *The mind's eye: Mind-wandering revealed by eye movement hidden Markov model*. Poster presented at the Annual Meeting of the Vision Sciences Society, Virtual meeting.
14. Kao, Y. W., **Lee, H. H.**, Goh, J. O. S., & Yeh, S. L. (2020). *Age-related differences in the statistical regularity of emotional faces*. Poster presented at the Annual Meeting of the Cognitive Neuroscience Society, Virtual meeting.
15. **Lee, H. H.,** Shih, C. Y., Chen, Z. L., Wu, A. Y., & Yeh, S. L. (2019). *Are you concentrating or mind-wandering? Electroencephalogram signals reveal differences in attentional states.* Poster presented at the Annual Meeting of the Society for Neuroscience, Chicago, IL.
16. **Lee, H. H.** & Yeh, S. L. (2019). *Blue-light effects on the gap effect.* Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
17. **Lee, H. H.,** Chen, Z. L., Chen, Y. T., Shih, C. Y., Beh, W. K., Wu, A. Y., & Yeh, S. L. (2019). *The heart never lies: Heart rate variability as an index of sustained attention.* Poster presented at the National Taiwan University-Kyoto University International Symposium for Cognitive Neuroscience, Taipei, Taiwan.
18. **Lee, H. H.** & Hsieh S. (2017). *The aging effect on time perception: An ERP study.* Poster presented at the 13th Asia Pacific Conference on Vision, Tainan, Taiwan.
19. **Lee, H. H.** & Hsieh S. (2016). *The relation between resting-state fMRI and stop-signal task performance in healthy middle-aged and elderly people.* Poster presented at the 55th Annual Taiwan Psychology Association Meeting, Tainan, Taiwan.

Awards and Honors

2026	**Postdoctoral Travel Grant Award** Issued by New York University
2026	**Trainee Professional Development Award (TPDA)** Issued by the Society for Neuroscience (SfN) *Society for Neuroscience, Washington, D.C. USA*
2025 – 2026	**Taiwanese Overseas Pioneers Grant** Issued by National Science and Technology Council (NSTC), Taiwan *PhD Candidate in Social Sciences*
2025, 2026	**Ted Coons Graduate Student Travel Award** Issued by Department of Psychology, New York University, USA *Vision Sciences Society, Florida, USA*

2023 **Graduate School of Arts and Science Dean's Conference Fund**
Issued by New York University, USA
Vision Sciences Society, Florida, USA

2021 – 2025 **Taiwan's Government Fellowship for Studying Ph.D. Abroad**
Issued by Ministry of Education, Taiwan
Area: Cognitive Neuroscience

2019 **College of Science Student Travel Award**
Issued by National Taiwan University
Summer School Program at University of California, Berkeley, USA

2019 **Scholarship for Students Attending International Academic Conferences**
Issued by Ministry of Science and Technology, Taiwan
Vision Sciences Society, Florida, USA

2017 **Scholarship for Students Attending International Academic Conferences**
Issued by National Cheng Kung University
Asia-Pacific Conference on Vision, Tainan, Taiwan

2016 **College Student Research Scholarship**
Issued by Ministry of Science and Technology, Taiwan
Project: The aging effect on time perception: An ERP study

2013 – 2016 **Academic Excellence Awards (awarded every school year)**
Issued by Department of Psychology, National Cheng Kung University
Top 5% GPA among the school year in Psychology Department

Teaching Experience

2026 Summer TA for Undergraduate *Perception*, New York University
2026 Spring TA for Undergraduate *Perception*, New York University
2025 Fall TA for Undergraduate *Cognition*, New York University
2025 Summer Instructor for Undergraduate *Cognition*, New York University
2025 Spring TA for Undergraduate *Perception*, New York University
2024 Fall TA for Undergraduate *Perception*, New York University
2024 Summer Instructor for Undergraduate *Perception*, New York University
2024 Spring TA for Undergraduate *Perception*, New York University
2023 Fall TA for Graduate *Perception*, New York University
2019 Spring TA for *Seminar in Graduate Institute of Psychology*, National Taiwan University
2018 Fall TA for *Introduction to Psychology*, National Taiwan University
2017 – 2018 TA for *Statistics in Psychology and Education*, National Taiwan University
2017 Spring TA for *Cognitive Psychology*, National Cheng Kung University

Invited talks

2026 Apr. *Modulation of perceptual sensitivity through visual adaptation and covert spatial attention.*
Public dissertation presentation, Department of Psychology, New York University.

2026 Mar. *How does visual adaptation affect contrast sensitivity? Evidence from TMS, psychophysics, and fMRI.*
Invited talk, Laboratory of Brain and Cognition, National Institutes of Health (NIH).

2026 Feb. *How does visual adaptation affect contrast sensitivity? Evidence from TMS, psychophysics, and fMRI.*
Invited talk, Reinhart Lab, Boston University.

2026 Jan. *The interaction between visual adaptation and visual attention.*
Invited talk, Project TYRA (Taiwanese Young Researcher Association).

2025 Jul. *Intrinsically photosensitive retinal ganglion cells and visual processing: ipRGCs beyond non-image-forming functions.*
Invited talk, Explorer of Perception and Attention Lab, National Taiwan University.

2025 Apr. *How do visual adaptation and covert spatial attention interact?*
Fourth-year research talk, Department of Psychology, New York University.

2024 Jul. *Linking differential adaptation effects on contrast sensitivity around the visual field to human primary visual cortex.*
Invited talk, Explorer of Perception and Attention Lab, National Taiwan University.

2023 Sep. *Adaptation shelters exogenous attention in early visual cortex: A TMS study.*
Second-year research talk, Department of Psychology, New York University.

2022 Sep. *Do exogenous attention and adaptation interact in early visual cortex?*
First-year research talk, Department of Psychology, New York University.

2021 Sep. *Blue-light effects on saccadic eye movements.*
Invited talk, Carrasco Lab, New York University.

2019 Mar. *Mind-wandering detection system with EEG and eye-tracking signals.*
Invited talk, PixArt Imaging Inc., Taiwan.

2018 May *Food information processed unconsciously under continuous flash suppression.*
Invited talk, Joint Conference of National Taiwan University, Peking University, and The Chinese University of Hong Kong.

Graduate Coursework

2024	Fall	Area Seminar: Perception & Attention
2023	Fall	Psychophysics
2023	Spring	Computational Cognitive Modeling
2023	Spring	Learning and Memory
2022	Fall	Mathematical Tools for Simulation and Data Analysis
2022	Spring	Functional Magnetic Resonance Imaging (fMRI) Lab
2022	Spring	Behavioral and Cognitive Neuroscience
2021	Fall	Perception
2021	Fall	Mathematical Tools for Neural and Cognitive Science

Ad-hoc Reviewer

Acta Psychologica, Attention, Perception & Psychophysics, Brain Sciences, Cognition, Cognitive Processing, Computers in Human Behavior Reports, European Journal of Neuroscience, Frontiers in Human Neuroscience, Frontiers in Psychology, IEICE Transactions on Information and Systems, Journal of Experimental Psychology:

Human Perception and Performance, Journal of Eye Movement Research, Neuroscience of Consciousness, PCI Registered Reports, PloS One, Psychological Research, Scientific Reports

Professional Affiliations

Asian-Pacific Conference on Vision (APCV), Cognitive Neuroscience Society (CNS), European Conference on Visual Perception (ECVP), Taiwanese Psychological Association (TPA), Society for Neuroscience (SfN), Vision Sciences Society (VSS)

Technical Expertise

R, MATLAB, Python, JASP, EEG (Neuroscan), fMRI (certified operator for Siemens Allegra 3T), Eye-tracker (EyeLink, Tobii), TMS (Brainsight, Magstim, MagVenture), Four-mirror stereoscope